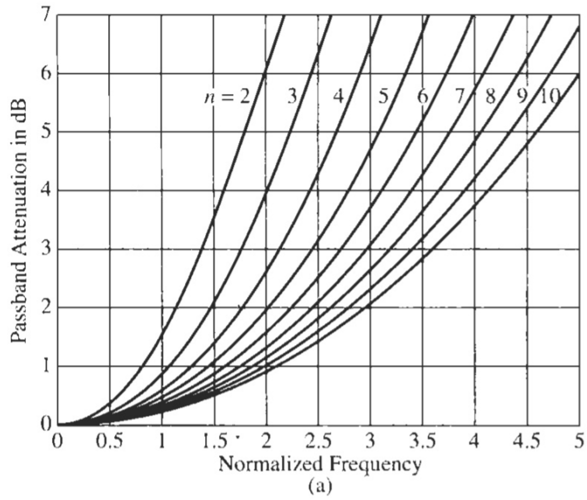
Filtro	Función de aproximación	Retardo constante	Desvío del retardo máximo	Atenuación máxima
E	Bessel	80 µs	5% @ 3 kHz	1 dB @ 2 kHz



- Con un filtro n=3 deberiamos llegar.

- $\text{cotgh}(s) = \dfrac{\cosh(s)}{\text{senh}(s)} = \dfrac{1}{s} + \dfrac{1}{\dfrac{3}{s} + \dfrac{s}{5}}$

$$= \frac{1}{s} + \frac{s5}{15 + s^2}$$

$$= \frac{s^2 + 15 + s^2 5}{s^3 + s15}$$

$$= \frac{s^2 6 + 15}{s^3 + s15}$$

- $\Omega_\omega = \dfrac{1}{\tau_0} = \dfrac{1}{80\mu S} = 12500 \text{ rad/s}$

- $T(s) = \dfrac{1}{\cosh(s) + \text{senh}(s)} = \dfrac{15}{s^3 + s^2 6 + s15 + 15}$

- $\omega_c = \dfrac{2\pi . 2kHz}{12500 \text{ r/s}} = 1 \text{ r/s}$

$$T(s)\Big|_{s=j\omega} = \frac{15}{-j\omega^3 - \omega^2 6 + j\omega 15 + 15} = \frac{15}{(15 - \omega^2 6) + j(\omega 15 - \omega^3)}$$

- $\omega_s = \dfrac{2\pi . 3kHz}{12500 \text{ r/s}} = 1,507 \text{ r/s}$

$$|T(1)| = \frac{15}{\sqrt{9^2 + 14^2}} = \frac{15}{\sqrt{277}} = 0,9012$$

- $\alpha_{max}(1) = -20.\text{Log}(|T(1)|) = 0,903 \text{dB}$

- $\phi(\omega) = -tg^{-1}\left(\dfrac{\omega 15 - \omega^3}{15 - \omega^2 6}\right)$

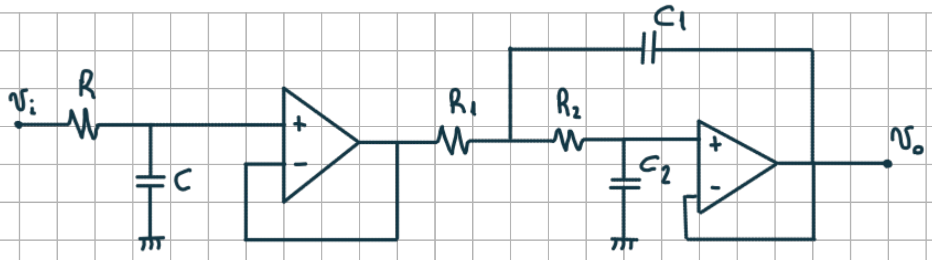
- $D(\omega) = \dfrac{-d\phi(\omega)}{dt} = \dfrac{225 + 45\omega^2 + 6\omega^4}{225 + 45\omega^2 + 6\omega^4 + \omega^6}$

- $\text{Error}(\%) = \left|\dfrac{D(\omega_s) - D(0)}{D(0)}\right| = \left|\dfrac{0,968 - 1}{1}\right| = 3,2\%$

$$\therefore T(s) = \frac{6,45944 \cdot 2,32219}{(s^2 + 3,67782 + 6,45944)(s + 2,32219)}$$

↳ $Q = 0,691$

- se propone el siguiente circuito (en función del Q)



• RC

$$\cdot T(s) = \frac{1/sC}{\frac{1}{sC} + R} = \frac{\frac{1}{RC}}{s + \frac{1}{RC}}$$

$$\cdot \Omega_\omega = \omega_o \quad \therefore \quad \frac{1}{RC} = 1$$

• con $C = 1$, $R = 1$

• Sallen Key

$$\cdot T(s) = \frac{\frac{1}{R_1R_2C_1C_2}}{s^2 + s\frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) + \frac{1}{R_1R_2C_1C_2}}$$

$$\cdot \Omega_\omega = \omega_o$$

$$\cdot 1 = \frac{1}{R_1R_2C_1C_2}$$

• Con $C_2 = 1$, $R_1 = \frac{1}{R_2C_1}$

$$\cdot \frac{1}{Q} = \frac{1}{C_1}\left(R_2C_1 + \frac{1}{R_2}\right)$$

$$\frac{1}{Q} = R_2 + \frac{1}{R_2C_1}$$

$$0 = R_2^2 - \frac{R_2}{Q} + \frac{1}{C_1}$$

• Corroboro que valor de C_1 da raices positivas

$$\frac{R_2^2}{Q^2} - 4 \cdot R_2^2 \cdot \frac{1}{C_1} > 0$$

$$R_2^2\left(\frac{1}{Q^2} - \frac{4}{C_1}\right) > 0$$

$$\frac{1}{Q^2} > \frac{4}{C_1}$$

$$C_1 > 4Q^2$$

$$C_1 > 1{,}909$$

→ Propongo $C_1 = 2{,}2$ (sabiendo valores comerciales)

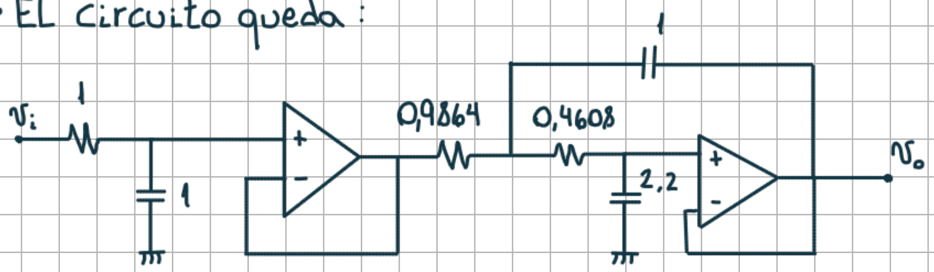
• Retomo: $\frac{1}{Q} = R_2 + \frac{1}{R_2 C_1}$

$\therefore R_2 = 0,4608$

• Retomo: $R_1 = \frac{1}{R_2 C_1}$

$R_1 = 0,9864$

• EL circuito queda:



• Desnormalización ($\Omega_\omega = 12500\ r/s$)

• RC

• M_i: $\Omega_\omega = 12500\ r/s \cdot 2,32219 = 29027\ r/s$

• Quiero un $C = 1nF$

• $C_r = \frac{C_n}{\Omega_\omega \cdot \Omega_z} \rightsquigarrow 1nF = \frac{1}{29027\ r/s \cdot \Omega_z}$

$\Omega_z = 34,45\ k\Omega$

• $R_r = R_n \cdot \Omega_z = 34,45\ k\Omega$

↳ Valor comercial $= 33\ k$

• Sallen key

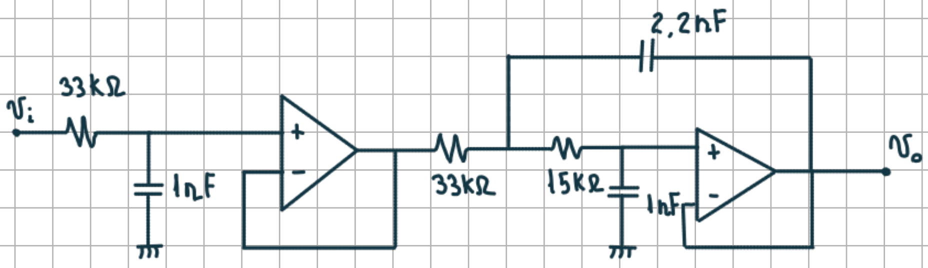
• $\Omega_\omega = 12500\ r/s \cdot \sqrt{6,45944} = 31769\ r/s$

• Quiero un $C_{2r} = 1nF$, $\therefore \Omega_z = 31477\ \Omega$

• $R_{1r} = 31\ k\Omega \rightarrow V_c = 33\ k\Omega$

• $R_{2r} = 14,5\ k\Omega \rightarrow V_c = 15\ k\Omega$

· Finalmente



· LTSpice